

Grade 11 – Term 1

C5 Practice Activity 5 with Solutions

Contents

1 Introduction

1

Evaluated criterion

C6. Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.

Evaluated criterion

C6. Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.

1 Introduction

Problems:

???. Continuous distributions, theoretical and empirical probabilities

???. Probability and relative frequency reference

Criterion C6 focuses on estimating interval probabilities from generated samples and comparing these empirical estimates with the theoretical probabilities predicted by a continuous probability distribution.

The comparison can be understood through two representations of the same distribution. The probability density function (PDF) represents the theoretical model as a smooth curve. A histogram represents the observations obtained from a finite generated sample. The theoretical probability of an interval is determined by the area under the PDF, while the empirical probability is estimated from the relative frequency of observations in the corresponding interval.

The normal distribution provides a useful example because its theoretical density has a smooth bell-shaped form, while a histogram generated from the same distribution produces a finite and irregular approximation to that shape.

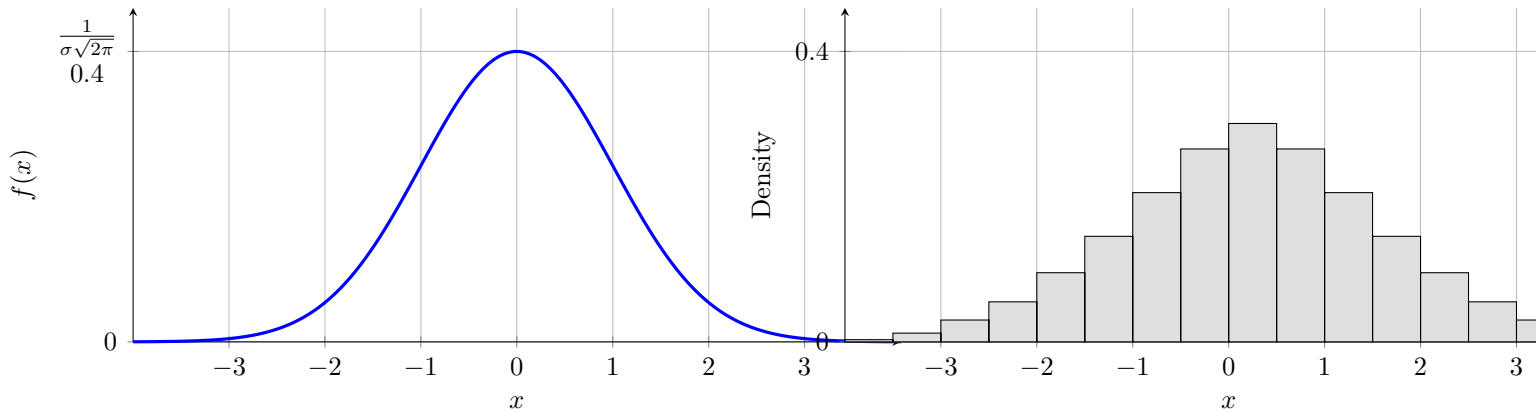


Figure 1: Theoretical probability density function of a normal distribution. The area under the curve over an interval represents the theoretical probability of that interval.

Figure 2: Histogram of a finite sample generated from the same distribution. The bars show the observed frequencies within the histogram bins.

$$P(a \leq X \leq b) = \int_a^b f(x) \, dx.$$

$$\hat{P}(a \leq X \leq b) = \frac{k}{n}.$$